Course Code: CSE4007	**Course Name:** Deep Learning	
Credits: 3 (2-0-2)	**Contact Hours:** 2 hour Theory & 2 hours lab per week	
Batch: 2023, 6th Sem **Academic Year:** 2025-26	**Semester Duration:** 19th Jan,2026 to 8th May 2026	
Course Faculty: 1. Dr. Hirdesh Pharasi [*hirdesh.pharasi@bmu.edu.in, 97755 50708 and office address*] 2. Dr. Manisha Saini [*manisha.saini@bmu.edu.in, 9999305497 and 22, IV Floor, E2 Building*]	**Course Coordinator:** **Name:** Dr. Manisha Saini **Email:** *manisha.saini@bmu.edu.in* **Office:** 62, IV Floor, E2 Building	

Aim of the course: The Deep Learning course aims to equip students with a strong foundation in deep learning concepts, architectures and applications. By combining theoretical understanding with practical implementation, the course prepares students to solve real-world problems in diverse domains such as image processing, text analytics, and audio processing using state-of-the-art deep learning techniques.

Course Overview and Context: Deep Learning is an elective course offered to 3rd year B.Tech students specializing in Data Science and Artificial Intelligence. The course provides an in-depth understanding of deep learning principles and methodologies, starting with a review of machine learning fundamentals and progressing to advanced topics such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and transformer-based attention models. Students will gain hands-on experience with Keras and Pytorch ,working on real-world applications like image classification, object detection, text classification, and deploying models on cloud platforms. With a strong focus on practical implementation, this course equips students with industry-relevant skills and prepares them for research-based careers or higher studies in artificial intelligence and data science.

Course Outcomes (CO): At the end of the course the students should be able to do the following:

	Course Outcome Statement	**Mapped Component**	**Associated Activities / Assessment**
CO1	Understand the underlying theoretical concepts behind different deep learning models including automatic feature learning and hierarchical feature representation.	Employability	→ Intensive theory sessions. → In-class tool demonstrations → Industry case studies etc.
CO2	Apply the knowledge to build different deep learning models capable of handling a whole variety of heterogeneous data including text, image, audio, etc.	Skill Development	→ Hands-on practical sessions in labs. → Experiential learning using frameworks (TensorFlow/PyTorch) → Mini projects implementing CNNs & transfer learning → Model training using GPU platforms (Google Colab, Kaggle NVIDIA etc.)

CO3	Analyze a given problem to design and deploy deep learning models, providing effective solutions to real-world problems.	Entrepreneurship, Skill Development	→ Group assignment designing a model deployment approach (API/Edge) → Case study showing how DL models can be applied for startup/industry needs

Prerequisite: if any

1. Basic knowledge of Python programming
2. Understanding of fundamental Machine Learning concepts

Topics of the course:

Topics	Number of Sessions (48)	Course Outcomes
Review of Machine Learning and Introduction to Deep Learning	3 (2 Th + 1 Lab)	CO1
Feed-Forward-Artificial-Neural-Network: Multi-Layered Perceptron – Optimization with Gradient Descent and Backpropagation; Variants of Gradient Descent – batch/minimatch/ stochastic-gradient-descent (SGD), Momentum, RMSProp, Adam. Weight initialization of an ANN	4 (2 Th + 2 Lab)	CO1, CO2
Using deep learning libraries – Introduction to Keras and Pytorch	4 (1 Th + 3 Lab)	CO1, CO2
Convolutional Neural Networks (CNN) – solving Binary and Multi-class Image Classification problems Object Detection from Images – RCNN, YOLO, SSD Image Segmentation - Semantic Segmentation with UNET Instance Segmentation with Mask-RCNN.	4 (1 Th + 3 Lab)	CO1, CO2, CO3
Processing sequence data with Recurrent Neural Networks (RNN), LSTM, GRU with applications to text classification.	4 (2 Th + 2 Lab)	CO1, CO2
Transfer Learning – Feature extraction and Fine-tuning Data Augmentation Embeddings – Word2Vec	6 (3 Th + 3 Lab)	CO1, CO2, CO3
Attention-based models and Transformers with applications	6 (3 Th + 3 Lab)	CO1, CO2
Autoencoders and Generative Adversarial Networks (GAN)	6 (3 Th + 3 Lab)	CO1, CO2

Applications of Deep Learning to Text, Image and Audio Processing	6 (2 Th + 4 Lab)	CO1, CO2, CO3
Deployment of the Deep Learning models on a cloud platform	4 (4 Lab)	CO3
Course Closure - Summary and Discussions	1 (1 Th)	_

CO/PO Mapping:

CO/PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1	1	1	1	2	2								1	1	2		
CO2	3	2	2		3								1	1	2		
CO3	1	2	3	2	2								3	3	1		

Learning Resources:

Textbooks:

1. Deep Learning by Ian Goodfellow and Yoshua Bengio and Aaron Courville, MIT Press, 2016.
2. Deep Learning with Python by François Chollet, 2nd Ed., Manning.
3. Dive into Deep Learning by Aston Zhang, Zachary C. Lipton, Mu Li & Alexander J. Smola, Cambridge University Press, 2023 .ISBN 978-1009389433.

Reference Books:

1. Dan Jurafsky and James H. Martin. Speech and Language Processing (3rd ed. draft).
2. Sherin Thomas & Sudhanshu Passi — PyTorch Deep Learning Hands-On, Packt Publishing, 2019

Video lectures/ MOOC:

Reference Links:
Due to the ever-changing nature of the web, current material that is freely available will be relied upon, including software documentation, blog posts, tutorials, and more. Links to the readings will be provided on a weekly basis in addition to the following online course:

Online Resource
1. Coursera Courses:

i) Deep Learning Applications for Computer Vision

https://www.coursera.org/learn/deep-learning-computer-vision

Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and

ii) Deep Learning

https://www.coursera.org/learn/introduction-tensorflow

iii) Image Classification with CNNs using Keras

https://www.coursera.org/projects/image-classification-cnn-keras

2. Deep Learning by Prof. Bhiksha Raj, Carnegie Mellon University

https://www.youtube.com/watch?v=7W2wR-VoMn0&list=PLp-0K3kfddPxRmjgjm0P1WT6H-gTqE8j9

3. CS231n: Convolutional Neural Networks for Visual Recognition

https://www.youtube.com/watch?v=vT1JzLTH4G4&list=PL3FW7Lu3i5JvHM8ljYjzLfQRF3EO8sYv

4. MIT Introduction to Deep Learning | 6.S191

https://www.youtube.com/watch?v=QDX-1M5Nj7s&list=PLTZ1bhP8GBuTCqeY19TxhHyrwFiot42_U

5. Popular blogs related to Deep Learning articles will be shared in the classroom.

e.g. Andrej Karpathy - Yes you Should Understand Backprop,

https://karpathy.medium.com/yes-you-should-understand-backprop-e2f06eab496b

6. Research Papers from arxiv (https://arxiv.org/)

Experiential Learning Component:

The project, as mentioned in the above table, is the primary experiential learning component of this course. Students will be given challenging real-life problems. They will be asked to build solutions by applying 5 suitable deep-learning algorithms. The students are also expected to implement and show the results of the proposed solution and perform a comparative analysis with different available algorithms. A separate will be conducted for evaluating the solution provided by each student. A student needs to spend approximately 15 to 20 hours to finish this assignment. The weightage of this component is 40%. The programming assignments (30% weightage) will help develop problem-solving abilities. Overall, the major focus of this course is on hands-on experiential learning. The total weightage to experiential learning is 70%.

Assessment Pattern: The final grade will be determined by the marks or grades earned during the project's phase-wise evaluations and the end-term assessment. Grading will be conducted using the relative grading method outlined in the university's academic regulations. To be eligible for grading, students must achieve a minimum of 40% of the total marks upon completing all assessments listed in the table below:

Evaluation Component	**Weightage (%)**	**Evaluation Schedule/Timeline**	**Rubrics/Remarks**	**CO mapping**
Assignment	20%	Continuous	Offline evaluation to be conducted during the LAB. Students will be asked to explain the solution with a demo followed by viva.	CO1, CO2, CO3
Mid-Semester Examination	20%	As per academic calendar/Date-sheet	Written Examination	CO2, CO3

Quiz	20%	April 3rd Week	MCQ knowledge and programming-based questions	CO1, CO2
End Term Project Evaluation	40%	As per academic calendar/Date-sheet	comprehensive Project report is to be submitted along with the coding solutions; this will be followed by Presentations, Demonstrations & Viva. Project-30%; Pre-submission-10%	CO1, CO2, CO3
– All evaluations will be based on the work presented by the students as well as the questions asked, or the problems given to code. – Cases of AI-generated code or plagiarism will be taken seriously and reported according to the university's policy on Unfair Means (UFM). It is essential that all work is original and adheres to academic integrity. – There is a mandatory requirement to upload the project to a public repository on GitHub.				–

Student Responsibilities:

- Attend lectures and do the work Lab Assignments as per instructions.
- Participate in the discussions/assignments held during classes.
- Check announcements on the LMS and emails regularly.
- Submit the assigned task on time.
- Regularly check marks on the LMS to ensure they are up to date.
- Participate in class and take necessary actions to grasp the material. Asking questions is encouraged.
- Communicate any concerns by speaking directly with the instructor.

Attendance Policy: Students are expected to attend classes regularly. Failure to follow the classes regularly and adhere to the expected attendance percentage will result in losing quiz/lab marks and a reduction of the grade as per the University's grading policy.

Recourse Examination Policy: In case a student fails the course, a one-time recourse is permitted as per the academic regulations of the University. Recourse is allowed only for the End Semester examination.

Make-up policy: No make-up exam will be conducted for unexcused absences. The faculty needs to be informed in advance in case the student is not going to appear for any evaluation component, and it is at the discretion of the faculty to sanction makeup for an evaluation component.

Behavior Expectations: No mobile phones and other distractive gadgets are permitted in the class.

Academic Dishonesty/Cheating/Plagiarism: Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.